

SkillSetu vs. SIH-2026-main: Role-Based Gap Analysis & Feature Roadmap

Complete Technical Blueprint: Automated Resume Onboarding • AI Certificate Credibility Engine • Server-Side Course Scout • 20-Feature Innovation Roadmap

Our Platform:

SkillSetu (Enterprise Production)

Compared Project:

SIH-2026-main (SkillBridge)

Privacy Standard:

100% Server-Side AI (No Student Keys)

Theme:

Smart Automation • Team SIGMA

Architecture Decision: Zero-Trust Client Model (No Student API Keys Ever)

Students are

never asked for personal API keys

. Requiring students to supply private keys damages trust and risks accidental leakage. All AI telemetry (Resume Ingestion, Skill Mapping, Course Scouting, and Certificate Verification) runs

100% on the platform's secure server infrastructure

using managed Gemini 2.5 Flash and local Python processors. The student experiences instant, frictionless automation at zero cost.

1. The Seamless 5-Step First-Login Onboarding Lifecycle

Onboarding Flow

How a new student transitions from initial login to verified dashboard readiness:

1 Initial Login

Student logs in via college credentials or OAuth. System detects first-time user flag (`is_onboarded: false`).

2 Immediate Resume Ingestion

System prompts drag-and-drop resume upload (PDF). Local server parses text in <100ms with DPDP PII masking.

3 AI Profile Synthesis

Server AI synthesizes preliminary profile: extracts skills, education, GitHub repos, and initial proficiency levels.

4 Interactive Review & Verification

Student reviews profile: adds missing achievements, projects, and uploads certificates for AI credibility audit.

5 Landing Page Launch

Student is redirected to Dashboard with populated match rings, verified badges, and personalized opportunities!

Strategic Innovation: Autonomous Server-Side AI Course Scout (No Student Keys Needed)

Core Architecture of Your Proposed Feature: An intelligent, continuous background agent running on the platform's backend infrastructure. It assesses student proficiencies, identifies emerging industry skill gaps, and continuously crawls accredited academic repositories for newly updated courses sorted by price tier.

1. Server AI Profiling

Runs on managed server AI. Parses GitHub commits, assessment answers, and project source code to diagnose true technical strengths vs. blind spots.

2. Real-Time Web Indexer

Background worker actively monitors accredited platforms (NPTEL, SWAYAM, MIT OCW, Coursera, AWS) for updated course releases.

3. Smart Price Tiering

Categorizes recommendations into 100% Free / Govt-Funded (₹0), Micro-Budget Boosters (<₹1,000), and Lab-Subsidized Institutional Grants.

4. Continuous Push Sync

As industry tech stacks evolve or student finishes modules, the system pushes dynamic alerts for newly matched free accredited modules.

2. Dedicated AI Certificate Credibility & Anti-Forgery Engine

Verification Architecture

The 3-Layer Credibility Pipeline for Student Certificates:

Layer 1: Multimodal Vision AI (Gemini 2.5 Flash Vision / LayoutLMv3) Scans document image for official issuing authority typography, recognized logo vectors, standardized course codes, and authorized digital signatures (NPTEL, Coursera, AWS, edX, IITs).	Layer 2: Metadata & Splice Detector (Python pikepdf & OpenCV) Inspects PDF XMP binary metadata. Flags documents modified by image editors (Photoshop, Canva, Illustrator), detects font layer splices (editing someone else's name), and catches altered dates.	Layer 3: QR & Registry Resolver (Python pyzbar + Webhooks) Decodes embedded QR codes or verification URLs (e.g. npTEL.ac.in/noc/Ecertificate) and queries the issuing academy's public endpoint to verify student name and completion grade.
Confidence Score	Status Assigned	Platform Action & Ledger Minting
≥ 90% Match	VERIFIED	Displays verified green check badge on student's public portfolio; mints SHA-256 certificate block to TrustLedger.
70% – 89% Match	PENDING_AUDIT	Added to profile as unverified self-claim until secondary manual or automated registry webhook confirmation.
< 70% / Tampered	REJECTED / FORGED	Certificate rejected with clear explanation: "Document failed metadata integrity check (Font layer splice detected)."

ROLE 1: STUDENT & ENGINEERING SCHOLAR

Capability	Present in SkillSetu (Our Website)	Status in SIH-2026-main (SkillBridge)
Onboarding & Resume Profiler	Immediate first-login drag-and-drop resume ingestion; server AI synthesizes profile and opens interactive edit wizard.	Completely Missing: No automated resume-to-profile parser; student must manually enter fields one by one.
AI Certificate Credibility Engine	3-layer verification (Vision AI + Metadata splice detection + QR registry query) before inclusion in portfolio.	Completely Missing: Allows unverified text inputs or raw files with zero authenticity or forgery checks.
Proctored Assessment & Anti-Cheat	Live Camera Proctor HUD + Fullscreen Lockdown + Window Blur & Tab-switch violation counter.	Completely Missing: Basic unproctored radio-button quiz; students can open tabs or copy answers with zero detection.
Public Verifiable Portfolio	Public vanity URL (/portfolio/:id) + Dynamic QR code + Live GitHub repository telemetry card (stars, forks, languages).	Completely Missing: Private authenticated profile only. No shareable public link, no dynamic QR code, no GitHub API integration.
Multi-Track Engineering Domains	Deep domain categorization across Software Development, Cloud & DevOps, Data Science & AI/ML, Embedded Systems, and Core Engineering.	Completely Missing: Single flat list of generic web tags with no deep multi-disciplinary domain categorization or skill ontologies.
Tiered Budget Course Filters	Interactive budget filter: Free/Govt Funded (₹0), Budget Boosters (<₹1,000), and Sponsored & Grant modules.	Completely Missing: Flat list of training courses with no price filters, subsidy tags, or grant eligibility indicators.

ROLE 2: INDUSTRY PARTNER & RECRUITER

Capability	Present in SkillSetu (Our Website)	Status in SIH-2026-main (SkillBridge)
Cryptographic Trust Verification	Recruiters can inspect candidate SHA-256 block hashes on the sovereign TrustLedger to verify score authenticity.	Completely Missing: Plain database integers in SQLite that anyone with backend access can alter undetected.
Proctor Integrity Telemetry	Displays real-time proctor integrity scores (e.g. 98% Integrity, 0 tab switches, 100% face presence).	Completely Missing: Zero proctor metrics; recruiters cannot tell if candidate cheated on technical screenings.
Live GitHub Code Review	Live repository telemetry inside the ATS candidate review card (top repos, star count, commit language breakdown).	Completely Missing: Relies solely on candidate text input descriptions with no real-time code verification.
Structured Role Requisitions	Supports posting Summer Technical Internships, 6-Month Co-Ops, Research Apprenticeships, and Graduate Trainee Openings with customized skill weights.	Completely Missing: Restricted to generic unstructured text job posts with basic fields.
1-Click Ledger Shortlisting	Shortlisting candidates mints a cryptographic SHA-256 transaction receipt anchoring candidate selection.	Completely Missing: Basic unhashed status change in a local database table with no tamperproof audit log.

ROLE 3: ACADEMICIAN & UNIVERSITY FACULTY

Capability	Present in SkillSetu (Our Website)	Status in SIH-2026-main (SkillBridge)
Automated Curriculum Harmonization	Uses AI to compare college course syllabi against active corporate job demands, revealing curriculum lag.	Completely Missing: Faculty dashboard only shows simple lists of created courses with no syllabus gap analysis.
Institutional Project Endorsements	Professors can digitally verify and sign capstone projects, minting university endorsement hashes on the TrustLedger.	Completely Missing: No project endorsement workflow or cryptographic faculty signature mechanisms.
Industry Collaborative Research & Lab Projects	Enables engineering professors and department heads to track industry-sponsored research, live corporate consultancy projects, and student lab co-development.	Completely Missing: Has zero infrastructure for faculty-industry research collaboration or corporate consultancy management.
Accreditation Audit Generation	Maintains auditable records of student mentorship hours and industry tie-ups required for NAAC & NBA reviews.	Completely Missing: No automated export or compliance reporting for academic accreditation bodies.

ROLE 4: INSTITUTIONAL ADMIN & REGULATORY AUTHORITY

Capability	Present in SkillSetu (Our Website)	Status in SIH-2026-main (SkillBridge)
National Workforce Heatmap	Interactive geographic heatmap of Indian states showing regional talent supply vs. corporate demand.	Completely Missing: Only shows elementary counter cards (Total Users, Total Jobs, Total Applications).
TrustLedger Node Monitor	Live telemetry monitoring distributed cryptographic verification nodes, block height, and health metrics.	Completely Missing: No ledger or blockchain architecture exists to inspect.
Regulatory Compliance Telemetry	Real-time compliance tracking against AICTE and UGC statutory accreditation benchmarks (NBA, NAAC, NIRF).	Completely Missing: No regulatory or policy alignment indicators.
DPDP Act 2023 Enforcement	Automatic PII scrubber sanitizing phone numbers, emails, and 12-digit Indian Aadhaar numbers before AI evaluation.	Completely Missing: Stores and processes unredacted student data with zero DPDP 2023 compliance.
Bilingual Language Engine	Dynamic topbar toggle switching the entire interface between English and Hindi (हिंदी) in real time.	Completely Missing: 100% hardcoded English with zero internationalization (i18n) framework.

STUDENT ROLE: 5 NEXT-GEN ADDITIONS**1. Immediate Resume Ingestion & Profile Wizard**

Prompts resume upload right after login, extracts full student profile via server AI, and opens interactive review wizard before redirecting to dashboard.

2. AI Certificate Credibility & Anti-Forgery Engine

Multimodal vision AI (Gemini Vision) + PDF metadata splice detector + QR registry resolver validating authentic credentials before ledger minting.

3. Autonomous Server-Side AI Course Scout

100% server-side AI (zero student keys required) that monitors accredited repositories (NPTEL, SWAYAM, MIT OCW) and filters by Free ₹0 vs. Budget tiers.

4. AI Voice & Video Mock Interview Simulator

Interactive AI avatar conducting role-specific technical and system-design interviews via voice, evaluating eye contact, clarity, and tech accuracy before live company drives.

5. Dynamic ATS Resume Scorer & 1-Click Keyword Tailor

Uploads PDF resumes against target job postings, generates an ATS compatibility score (0–100%), and suggests exact missing keyword improvements inside a slide-over drawer.

RECRUITER ROLE: 5 NEXT-GEN ADDITIONS**1. In-Browser Proctored Coding & Engineering Sandbox**

Interactive coding environment (Python, TypeScript, C++, SQL, or Docker configurations) with real-time proctored webcam recording, code execution telemetry, and zero local setup.

2. Predictive Candidate Offer Acceptance & Retention Index

AI analytics evaluating candidate travel radius, past interview conversion history, and stipend fit to predict probability of offer acceptance and 1-year retention.

3. Tamperproof Smart-Contract Letter of Intent (LOI) Minting

One-click generation of cryptographically signed, verifiable Letters of Intent (LOIs) anchored on the TrustLedger upon selecting a candidate.

4. Blind Resume Screening (Unbiased Merit Mode)

Toggle that temporarily masks candidate name, gender, college tier, and photo during technical rounds—ensuring 100% bias-free, meritocratic evaluation.

5. Automated Bulk Campus Interview Auto-Scheduler

Integrates with Google Calendar, Microsoft Teams, and Zoom to auto-schedule hundreds of technical and HR rounds across colleges, eliminating recruiter scheduling overhead.

ACADEMICIAN ROLE: 5 NEXT-GEN ADDITIONS**1. AI Curriculum Diff Engine (Syllabus Re-Harmonizer)**

Professors upload existing university syllabus PDFs; AI compares them with 6 months of live recruiter postings to output an exact diff of missing vs. outdated topics.

2. Corporate Consultancy & Sponsored R&D Exchange

A dedicated marketplace where industries post live technical bottlenecks and university professors can submit bids to provide funded corporate consultancy.

3. One-Click NAAC / NBA Accreditation Dossier Generator

Instant compilation of student internship attendance records, industry mentorship hours, MoUs, and verified placement statistics formatted for accreditation bodies.

4. Industry Guest Lecture & Workshop Scheduling Portal

Allows universities to invite verified corporate engineering leads and industry architects for accredited guest lectures and hands-on laboratory workshops.

5. Collaborative Capstone Industry Co-Mentorship Hub

Enables corporate senior architects and engineers to co-mentor final-year college capstone projects alongside faculty, bridging theoretical research with production reality.

ADMIN & REGULATORY ROLE: 5 NEXT-GEN ADDITIONS**1. Predictive National Skill-Shortage Early Warning System**

Forecasting dashboard modeling which technical domains will suffer acute talent deficits across India over the next 1–3 years (e.g. Semiconductor VLSI, Quantum Computing, Cloud Security, Edge AI).

2. Academic Bank of Credits (ABC) & DigiLocker Gateway

Directly syncs verified internship hours, proctored test badges, and training credits into the student's official DigiLocker ABC account under NEP 2020.

3. Fraudulent Job Posting & Scam Recruiter Detector

Autonomous fraud filter scanning job postings for ghost companies, suspicious fee requests, or fake internship drives, blacklisting illicit accounts.

4. Geofenced Regional Skilling Grant Optimizer

AI model recommending where central government skilling subsidies should be allocated based on regional deficit data (e.g. deploying IoT labs in Tier-3 colleges).

💡 **Pitch Defense for SIH Judges:**

"Judges, while conventional approaches like SIH-2026-main remain static single-stream job boards with basic forms, SkillSetu operates as a sovereign national infrastructure. We solve resume fraud using a **Sovereign SHA-256 TrustLedger**, eliminate cheating via **Live Camera AI Proctoring**, protect student privacy with **DPDP Act 2023 Aadhaar masking**, connect Students, Educators, and Recruiters in a **bilingual Hindi/English** smart automation ecosystem, and feature an **Autonomous Server-Side AI Course Scout & Certificate Credibility Engine** that continuously verifies student credentials and scours accredited repositories like NPTEL and SWAYAM to keep student skills permanently aligned with industry demand."